Exploring the Nature of Chinese Characters from Zhao Yuanren's Perspective on Zi （字）and Ci（词）

——A Study of Chinese Characters from the Perspective of New Language state

Ryan Kangjie CHANG

The College of Chinese Language and Literature, Hunan University

Zhejiang Institute of Mechanical and Electrical Technician

Abstract:

This paper, based on Yuen Ren Chao’s theory of Zi（字 or Chinese character）and Ci（词 or Word）, explores the fundamental features and functions of Chinese characters. As a founding figure of modern Chinese linguistics, Chao’s unique insights have profoundly influenced the understanding of Chinese characters and the Chinese language. This study aims to uncover the essence of characters and words in Chinese linguistics and proposes a new theoretical framework — the “New Linguistic State” theory — which posits that language consists of a language generation mechanism and a language storage-transmission mechanism. By reviewing Chao’s perspectives, this paper argues that the theory of Chinese characters from the perspective of New Linguistic State can clarify existing misunderstandings about Chinese characters and provide a universal theoretical framework for various written symbol systems, thereby advancing the development of universal Chinese character studies.

Keywords: Zhao Yuanren,Zi（字）and Ci（词）, Language state， New Language State

1. Introduction

In the development of modern Chinese linguistics, Zhao Yuanren (1892–1982) is recognized as one of its founding figures. His research on the relationships among Chinese characters, Zi (字), and Ci (词) has had a profound influence on both domestic and international academic communities. Particularly, Zhao’s perspectives on distinguishing the concepts of Zi and Ci were remarkably forward-thinking. He not only identified the distinctions between Zi and Ci in Chinese linguistics and the sociological implications of Zi, but also recognized their inherent differences within the linguistic framework. His analyses illuminated the relationship between Chinese characters and the language system, offering a novel perspective for understanding the functions and distinctions of written language symbols.

In current research, the conceptual definitions of Zi and Ci remain ambiguous. The distinction between the "form-sound-meaning" unity of Chinese characters and the separateness of words in Western languages has yet to be systematically explained. Additionally, the study of a universal theoretical framework for written symbol systems remains underdeveloped. These issues have constrained further understanding of the essence of Chinese characters and their roles within linguistic systems.

Building upon Zhao Yuanren's theoretical foundation, this paper analyzes the core ideas of his views on Zi and Ci, proposing the New Language State Theory to address these challenges. The central contribution of this theory lies in redefining Zi and Ci from dual perspectives: the mechanisms of language generation and the mechanisms of linguistic storage and transmission. It reveals their essential roles in the Chinese language system. Furthermore, based on this analysis, the paper offers a universal theoretical framework for cross-linguistic symbol systems, thereby expanding the applicability and significance of research on Chinese characters.

By introducing the New Language State Theory, this study aims to explore the unique functions of Chinese characters and their critical position in the modern Chinese linguistic system, paving new pathways for the study of Chinese characters.

2. Analysis of Zhao Yuanren's Views on Zi（字）and Ci（词）

Zhao Yuanren's theory of Zi (字) and Ci (词) is primarily reflected in three significant works: The Problem of the Chinese Language (1959), A Grammar of Spoken Chinese (1968), and The Concept of Chinese Words: Their Structure and Rhythm (1975). These three works, spanning approximately 16 years, represent the mature scholarship of Zhao's later years and provide a relatively comprehensive depiction of the evolution of his thoughts on Zi and Ci. Through these works, Zhao's understanding of Zi can be observed to progress from morphemes to sociological words, and ultimately to word-syllables[1].

2.1 The Evolution from Morphemes to Word-Syllables

In The Problem of the Chinese Language, Zhao equates Zi with morphemes. By the time of A Grammar of Spoken Chinese, he redefined Zi as “sociological words.” Finally, in The Concept of Chinese Words: Their Structure and Rhythm, Zhao adopted Peter A. Boodberg’s notion of phonosemanteme, a phonetic and semantic unit below the level of a word in Chinese, and referred to it as Zi. When more convenient, he termed it “word-syllable” (Zhao, 2002:894), emphasizing that a Zi is both a syllable and typically carries meaning.

Zhao’s view of Zi evolved from morphemes to sociological words and ultimately to word-syllables, reflecting a systematic understanding of the core attribute of Zi in Chinese as a combination of sound and meaning.

2.2 The Refinement of the Definition and Standards of Ci

Zhao’s theory of Ci (词) also underwent a process of refinement, shifting from a formalist to a functionalist perspective. Initially, he used the criterion of independent usage to define Ci, proposing four methods for identification: morpheme order, tone, phonological variation, and word class. In A Grammar of Spoken Chinese, he introduced the concept of “syntactic words/sentence-building words,” emphasizing the relational syntactic chain within sentences and opposing the strict definition of Ci as the “smallest independently usable linguistic unit.” By the time of The Concept of Chinese Words: Their Structure and Rhythm, Zhao explicitly noted the fundamental differences between Zi in Chinese and words in English. He explained that his insights were informed by modern linguists’ growing awareness of these distinctions. Scholars such as Li Jinxi consciously linked Chinese linguistic units to the English

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1 Shi Youwei's "A Preliminary Exploration of Zhao Yuanren's 'Character-Word' Thought" Presented at the International Symposium on Zhao Yuanren's Linguistic Thought, August 21-22, 2023.

“word,” which led to the concept of “structural words.” This concept emphasized formal criteria for defining word boundaries while recognizing the divergence between Chinese and Indo-European languages. Throughout this progression, Zhao consistently underscored the importance of formal criteria, consciously downplaying the role of semantic factors in determining Ci.

Zhao’s evolving perspective on Ci reflects a deepening focus on functionality, balancing formal criteria within syntactic chains and revealing the essential differences between Ci in Chinese and words in Indo-European languages. This provided a new perspective for defining linguistic units.

2.3 Dual-Track Characteristics and Theoretical Implications

Zhao Yuanren’s theory of Zi and Ci demonstrates a unique dual-track characteristic——the parallel development of grammatical and sociological dimensions, as well as the coexistence of sentence-building functionality and rhythmic features. His work significantly influenced subsequent research, notably inspiring Xu Tongqiang’s “Zi-centrism” theory. Zhao’s academic attitude was marked by caution, maintaining an open stance on unresolved issues, leaving conceptual space for critical examination, and engaging critically with Western linguistic theories. This cautious and open approach reflected his humility toward complex linguistic phenomena and created broad opportunities for future exploration. While some aspects of Zhao’s views remain ambiguous, this ambiguity underscores his respect for the intricacies of language.

An analysis of Zhao Yuanren’s perspective on Zi and Ci reveals his gradual adoption of Western academic thought. His ideas provided critical inspiration for Xu Tongqiang’s “Zi-centrism” theory. As Shi Feng (2023) observed, Xu Tongqiang’s “Zi-centrism” is essentially a “morpheme-centrism.” In this context, the author argues that “Zi-centrism” could also be interpreted as the sociological Zi in Chinese. The debate over linguistic centrality in Chinese has been a focal point among domestic scholars, especially against the backdrop of national efforts to promote “neglected ancient studies,” such as archaic scripts. This provides scholars of language and writing an opportunity to explore the essence of Chinese written characters as a system of written symbols.

Therefore, this paper aims to investigate the essence of Chinese characters from Zhao Yuanren's perspective on Zi and Ci, while discussing the distinctions between Zi, Ci, and the sociological Zi in Chinese linguistics.

3. Exploration of the Essence of Chinese Characters

In traditional views of writing, it is commonly accepted that "writing is a symbol that records language." Therefore, before discussing the essence of Chinese characters, it is essential to address the concepts of symbols and the relationship between symbols and language.

Regarding symbols, Zhao Yiheng(2012) provides a series of definitions: "A symbol is a perception that carries meaning: meaning must be expressed through symbols, and the purpose of symbols is to convey meaning. Conversely, meaning that does not require symbols to express does not exist, and there are no symbols that do not convey meaning."（Zhao Yiheng 2012， pp1）"Meaning is the potential of a symbol to be interpreted by another symbol, and interpretation is the realization of meaning." （Zhao Yiheng 2012，pp2）"Meaning must be interpreted through symbols, and symbols are used to explain meaning. Conversely, there is no meaning that can be explained without symbols, and no symbol that does not explain meaning." （Zhao Yiheng 2012，pp2） "A symbol is meaning, and without symbols, there is no meaning. Semiotics is the study of meaning." （Zhao Yiheng 2012，pp3）The discussion in semiotics actually addresses the long-standing paradox in philology that "writing is a symbol that records language, and since symbols have meaning, writing has meaning."

In examining the relationship between symbols and language, we must recognize the following points[2]: (1) the distinction between symbols and entities outside of

[2] This part is the author's opinion obtained from Professor Peng Zerun of Hunan Normal University in private. It cannot represent all of Teacher Peng's views. If there are any errors or omissions, it is the author's responsibility and has nothing to do with Teacher Peng.

symbols. Both are entities that combine form and content; however, symbols treat entities outside of themselves as their content. (2) The two forms of language symbols. The fundamental form of a language symbol is phonetics; however, phonetics is not a symbol. The true symbol is oral language. The transformation of oral language from auditory to visual representation occurs through written symbols, wherein characters are not symbols but merely forms of symbols. (3) The "enhancement" of written language over spoken language. Written language, after converting oral language into a visual form, can be modified to become more concise and fluid, allowing for repeated and non-linear reading.

Discussing the essence of writing and Chinese characters based on the relationship between symbols and language is the core issue in writing studies. We will refer to the discourse of scholars in this field to address related issues concerning writing and Chinese characters. Qiu Xigui argues that when defining phonetic writing, one must examine what components of language the letters within a character represent. Consequently, when characterizing Chinese characters, one should not focus on what a character represents, but rather on how the components within a character relate to aspects of language (QiuXigui2020).

The relevant theories put forth by Qiu Xigui can be summarized as follows: (1) A character is a symbol that records a morpheme, where one morpheme is recorded by one character. If two symbols record different morphemes, then they are distinct characters; if they record the same morpheme, they are the same character. (2) A character is the smallest written symbol, represented as a square character. If two square characters have identical shapes, they are the same character; if their shapes differ, they are distinct characters.

It is evident that Qiu's discussion primarily analyzes the unique writing system of Chinese characters, rather than writing in the linguistic sense. The first point mentioned above asserts that writing is a symbol that records morphemes, indicating that characters carry morphemes. This visual representation of meaning is artificially assigned. Writing is a written language symbol, and the reason I hold this view is that the fundamental form of a language symbol is its phonetic representation, with true symbols being oral language. The transformation of oral language from auditory to visual through written representation forms written language, which is the true symbol. At this point, writing corresponds to phonetics and is one of the forms of language. This perspective aligns with Zhao Yuanren's view of "characters" in a sociological context, whereby written language carries meaning and acts as a symbol.

Wang Ning(2014) discusses Chinese characters from a functional perspective, proposing two states: usage state and storage state. We understand that Chinese characters are in a usage state when recording Chinese texts, possessing verbal meanings that may slightly fluctuate in terms of character count, frequency, and coverage with changes in social language usage. Conversely, characters in a storage state exist within dictionaries, detached from their original linguistic environment and categorized according to lexicographical criteria, displaying attributes of form, sound, and meaning. Wang notes that processing characters in a storage state is more complex than processing those in a usage state. Regarding the relationship between characters and words, she asserts that a word is the smallest independently usable unit of meaning in language, combining sound and meaning, while Chinese characters record these words. Given that ancient Chinese primarily utilized monosyllabic words, there was often no clear distinction between characters and words, with "name" or "character" commonly used to denote what we understand today as "words."

In this context, the "character(字)" we discuss aligns with Zhao Yuanren's assertion that "in the Chinese conceptual framework, 'characters' are the central theme," consistently appearing in the form of "form + word/morpheme." Through comparative analysis, we can observe that Zhao Yuanren's perspectives on characters and words not only delve into the essential characteristics of Chinese characters but also provide a valuable theoretical foundation for subsequent linguistic research. For instance, the "characters" referred to by Wang Ning（2014）, which are aligned with Qiu Xigui's first property of "characters," are all linguistic units that record language in the form of "form + word/morpheme." Words/morphemes are a combination of sound and meaning, which, in turn, are closely linked to form, leading to a high degree of unity between form, sound, and meaning. This "form + word/morpheme" structure corresponds to Zhao Yuanren's notion of "sociological characters," which is equivalent to the concept of "word" in English. Although scholars like Qiu Xigui and Wang Ning discuss the relationship between characters and words from different angles, their work is undeniably influenced by Zhao Yuanren's theories.

I will refer to Zhao Yuanren's concept of "sociological characters," as well as Wang Ning's all-encompassing notion of "characters" and Qiu Xigui's first property of "characters" （Qiu Xigui2020）,as the "language state."

4. Research on Chinese Characters from the Perspective of the New Language State Theory

The "New Language State Theory" builds upon Zhao Yuanren's perspectives on Zi (character) and Ci (word), aiming to reassess the symbolic functions and linguistic essence of Chinese characters through the dual lenses of language generation mechanisms and storage-transfer mechanisms. At its core, the theory introduces the fundamental unit of "language state," a product of the integration of structural states and cognitive states, characterized by uniqueness and generativity.

4.1 The Concept of Language State

Our concept of the language state (Language State) is closely related to the grammeme introduced by Lu Chuan（2002）. He explains that the auditory sensitivity unit for the Han people is the "syllable," which means that the basic unit of Chinese should be the "monosyllable." Furthermore, Chinese possesses the function of "signifying meaning through character form," so the basic unit of Chinese must also consider "visual imagery." Given that Chinese characters often have multiple meanings, the basic unit should be positioned at "one meaning of a Chinese character." He provided his own view on the grammeme, as shown in the following diagram:

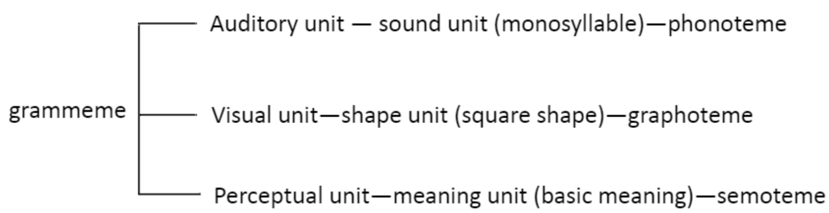

Figure 1. Lu Chuan's (2002) Conceptual Diagram of the Grammeme

Lu Chuan's related discussion introduces the concept of the grammeme. However, his analysis does not clarify the nature of Chinese characters or the essential nature of the basic unit of Chinese characters—the grammeme. This is partly due to the current lack of a comprehensive understanding of the nature of Chinese characters. Zhao Yuanren, in contrast, also provided an explanation of the grammeme by referring to morphemes as grammemes. However, Zhao's interpretation differs from Lu's, which is why I refer to it as the language state (Language State).

The language state (Language State) refers to the linguistic unit that combines the "form" and "word/morpheme" of a character. In this sense, we recognize that a word is a combination of sound and meaning, but are sound and meaning part of the language mechanism? In fact, Deng Siying(2019), in citing several formal syntacticians, raised a similar question: Chomsky (2000) and Hauser, Chomsky, and Fitch (2002) argue that sound and meaning are not inherently part of the language mechanism. Rather, it is the product of the processing of sound and meaning through various departments of the language mechanism that constitutes part of the language mechanism, and this product is ultimately stored in the lexicon of the language mechanism.

The "language mechanism" referred to here is essentially the language mechanism in generative linguistics. According to the minimalist program, the language mechanism consists primarily of the "cognitive system." The cognitive system includes both the "computational system" and the "lexicon" of human language. The lexicon stores words and marks their characteristics, functioning as the department responsible for word formation—morphology. The computational system, also referred to as the syntactic component or "narrow syntax," is responsible for operations(Deng Siying 2019,pp11-12).

The specific language mechanism is depicted in the following diagram:

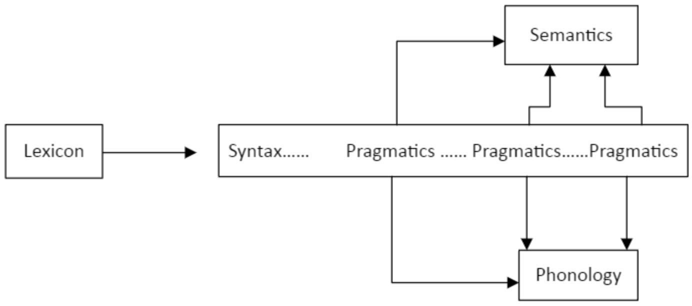

Figure 2. Deng Siying's (2019) Diagram of the Language Generation Mechanism

Thus, the product of syntactic operations is what Zhao Yuanren termed "structural words," which also form part of the "word/morpheme" section of the language state (Language State), combining both form and morpheme. These products are stored in the lexicon. This view differs significantly from traditional Chinese language research. According to the latest Distributed Morphology (DM) theory, the lexicon is eliminated, and word formation is understood as a direct product of syntactic operations on computational atoms. This theory, in my view, aligns more closely with traditional Chinese studies. Hu Xuhui(2022) provides a detailed explanation of word formation in the DM model:

If the lexicon does not contain words and is not responsible for generating words for syntactic operations, then what exactly is stored in the lexicon? Distributed Morphology (DM) suggests that the lexicon contains the following:

List 1: Abstract morphemes[3] and roots (i.e., syntactic terminal elements and the objects of syntactic operations).

List 2: Vocabulary items (phonological realizations of syntactic derivations).

List 3: Encyclopedic knowledge (semantic realizations of syntactic derivations).

The elements from these lists are retrieved at different stages of language

[3] Morphemes here are not traditional structural morphemes.

structure generation. Based on this, we can obtain the T-model diagram of Distributed Morphology (DM):

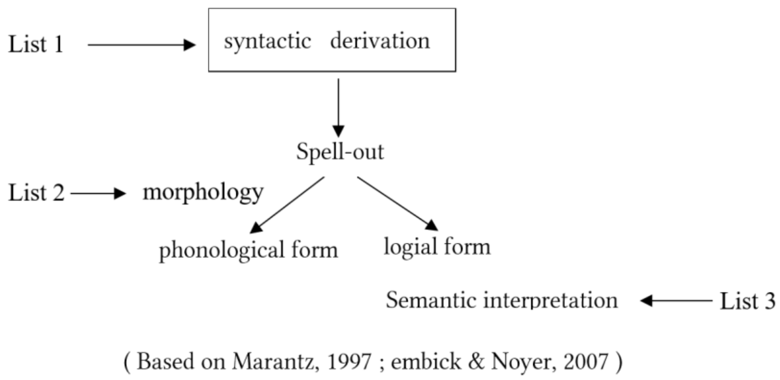

Figure 3. T-Model of Distributed Morphology (DM)

According to this diagram, I argue that in the DM model, the language generation mechanism produces language sequences arranged under specific conditions, which correspond to what is traditionally referred to as "words." Zhao Yuanren termed these "structural words," and we refer to them as cognitive states. In contrast, the formal features of written language are the products of a specific rule-based arrangement of forms, traditionally called "characters." By analogy with the cognitive state, we can call this the structural state. Based on this, we propose a new definition of the language state (Language State): a language state is a linguistic unit formed when the structural state and the cognitive state activate and use their referential functions under specific conditions to store and transmit linguistic information.

Furthermore, building on the work of Chang Kangjie(2024), we propose the theory of the New Language State, which holds that the language mechanism includes the Language Generation Faculty and the Language Storage-Transfer Faculty. To avoid confusion with formal syntactic mechanisms, we refer to the language mechanism in the theory of the New Language State as the language generation state.

4.2 Two Mechanisms for Generating Language States

The Theory of New Language State comprises two components: the Language Generation Faculty and the Language Storage-Transfer Faculty.

Language Generation Faculty.The Language Generation Faculty is primarily responsible for producing spoken language. Its symbols correspond to "words" in linguistics and are in charge of the integration of phonetics and semantics, as well as syntactic generation, thereby producing a Cognitive State of language. This mechanism generates linguistic expressions in the form of phrases and sentences that carry meaning and can be understood by speakers.

Language Storage-Transfer Faculty.The Language Storage-Transfer Faculty refers to the process by which cognitive linguistic encoding sequences (hereafter referred to as Cognitive State Sequences) need to be naturally transmitted through a visual storage form. This requirement stems from the essential function of language as a tool for communication.

Historically, due to technical limitations or the excessive flexibility of lingua franca, the Cognitive State Sequences generated by the Language Generation Faculty could not be effectively stored or transmitted. This limitation hindered communication. Therefore, it became necessary to utilize a form that allows for the storage and transmission of these cognitive states and encoding sequences.

In practice, the written language produced by the Language Storage-Transfer Faculty corresponds to "characters" in linguistics. We refer to this as the Structural State or Language Form State. In this process, the Language Storage-Transfer Faculty converts cognitive states into written forms, thereby enabling the storage and transmission of language. The written form ensures that linguistic expressions can be preserved over time and shared across different contexts, thus fulfilling the communicative function of language.

4.3 Language Storage-Transfer Faculty

The Language Storage-Transfer Faculty consists of three parts: Encoding and Storage, Transmission, and Decoding and Storage Retrieval.

Encoding and Storage.The Encoding and Storage process can be divided into two stages: the Generation and Storage Stage and the Encoding and Recording Stage.The Generation and Storage Stage operates in parallel with the lexicon (morphology) in the Language Generation Faculty.The Encoding and Recording Stage runs parallel to the syntactic system in the Language Generation Faculty.During the Generation Stage on the Visual Plane, the Cognitive State derived from syntactic processes is mapped to visual structural features under certain conditions. This mapping activates a referential function, resulting in the formation of what we call a Structural State. It is important to note that the Cognitive State is a pre-existing entity generated by natural syntactic processes, while the visual structural features are encoded representations based on certain characteristics of the Cognitive State. Both the Cognitive State and the visual features already exist independently. Therefore, the visual plane encoding process is essentially a rule-based shape generation process.We define this visual encoding as the Structural State: a form generated by mapping visual features according to certain rules. To illustrate this concept, the Structural State can be metaphorically compared to a sealed opaque bottle. During the encoding process, the Cognitive State is placed inside this opaque bottle for storage. The resulting filled opaque bottle represents an overlapping state of Structural State (the bottle) and Cognitive State (its content). This combined state is what we refer to as the Language State, where form, sound, and meaning coexist as a unified whole.

Encoding and Recording Stage.The Encoding and Recording Stage is essentially the process of ordering the filled opaque bottles (Language States) in accordance with the syntactic system. This sequence is referred to as the Language State Sequence.

Transmission.The Transmission Stage involves passing on the ordered sequence of filled opaque bottles (Language State Sequence) from one individual or context to another.

Decoding and Storage Retrieval.During the Decoding and Storage Retrieval process, the ordered opaque bottles (Language State Sequence) are decoded to reveal both their Structural State (the bottle) and the Cognitive State (the content inside).

I think in the study of ancient scripts, what we observe is essentially the opaque bottle (the Structural State). The process of interpreting ancient characters and conducting comparative etymological studies is, in fact, a process of decoding the stored language states. The author argues that the essence of traditional philology lies in the study of Language States.

Similarly, the debate over the fundamental unit of modern linguistics can be understood through the lens of the Decoding and Storage Retrieval process. When scholars view characters (written symbols) as the core unit of analysis, they are, in reality, focusing on decoding Structural States rather than the underlying Cognitive States.

This perspective aligns with the proposal made by Professor Xu Tongqiang regarding the “Character-Based Approach” in Chinese linguistics. The author suggests that what Xu actually advocated was a “Language-State-Based Approach”, which better captures the underlying nature of linguistic encoding, storage, and retrieval.

4.4 Characteristics and Generation Process of Language State

We propose the concept of the Language State, which refers to a linguistic unit formed when the referential functions of the structural state and the cognitive state are activated and utilized under specific conditions. The Language State possesses two primary characteristics: uniqueness and generativity.

Uniqueness refers to the fact that the semantic interpretation of encyclopedic knowledge within a logical structure is unique, which imparts the Language State with uniqueness. Since the combination of abstract morphemes and roots is unified, and the Theory of New Language State posits a one-to-one correspondence between the structural state and its mapped cognitive state, any change in the cognitive state, regardless of whether the structural state remains the same, results in the formation of a new Language State.

Generativity refers to the process by which different components of the Language State Mechanism in the human brain generate language state features. These language state features are then used by the Language Generation Faculty and the Language Storage-Transfer Faculty to produce the structural state (Structure State) and the cognitive state (Cognitive State). When the cognitive state and the structural state are activated by the Language State Mechanism, they combine to form the

Language State.

We will illustrate the practical application of the New Language State Theory using the term “人民” (the people).

the term " rén mín 人民" (people) is a language state. Analyzing " rén mín 人民" into " rén 人" (person) and "mín 民" (people), these elements seem like structural states and syllables. Based on generativity, they map out different meanings, forming new cognitive states and sub-language states ⟦rén 人⟧ and ⟦ mín 民⟧. This illustrates the concept of the language state as the smallest and freest linguistic unit. Zhao Yuanren's concept of the grammeme and grammeme group aligns with our language state theory.

Comparing the language states " rén jiān 人间" (the human world) and " rén mín 人民" (people), the morphemes " rén 人" and "jiān 间," "rén 人" and "mín 民" can be decoded using cognitive state theory. However, these morphemes consist of three structural states. Importantly, the sub-language state ⟦rén 人⟧ in " rén jiān 人间" differs from that in " rén mín 人民" due to different cognitive states.

Based on uniqueness, we can identify language states by checking if the structural and cognitive states match. Note that the lexical meanings of Chinese cognitive states are explicit, while functional meanings of abstract morphemes are implicit. This overlap occurs in the storage phase of the Language Storage-Transfer Faculty after language generation. Lexical and functional meanings are combined in words through storage encoding. The resulting lexical entry, or "dictionary word," is a product of human creation, not equivalent to the lexicon. Activated structural states generate language states, recording lexical items during language expression. This is the essence of Wang Ning's (2014) distinction between the "use state" and "storage state" of Chinese characters.

Regarding the overlap of lexical and functional meanings, each word’s functional meaning is implicit, while functional words’ roles in sentences emerge after processing by the Language Generation Faculty. During the cognitive decoding process, the receiver extracts these implicit features, a process known as grammaticalization.

4.5 Biological Evidence for the Language State Mechanism

Daniel Mitropolsky and Christos H. Papadimitriou(2023)provided a biologically plausible explanation of how language is acquired in the infant brain, not merely processed by a mature brain, by simulating a biologically credible language organ. Their discussion of the Language Organ aligns well with the concept of the Language State Mechanism.

Mitropolsky and Papadimitriou(2023)offer the example of "dog" (as shown in the figure below). In this model, the VISUAL and MOTOR areas are two fundamental context regions, although a semantic area would also encompass additional context regions. In the motor perception part (MOTOR area), "the dog moves," and in the visual perception part (VISUAL area), we perceive a static "dog." Both are inputs to the language environment and are essentially roots within the framework of Distributed Morphology (DM). The LEXV and LEXN that combine with these roots represent abstract morphemes. Once combined with phonological information, they form vocabulary items, completing an operation within the Language Generation Faculty.

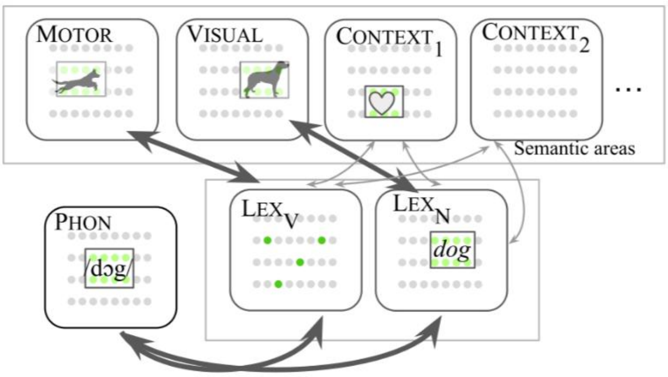

Figure 4. Language acquisition model by Daniel Mitropolsky and Christos H. Papadimitriou (2023).

It is crucial to note that Mitropolsky and Papadimitriou's model validates the possibility of the Language Generation Faculty, while their analysis focuses on language acquisition. However, for the full expression of a language state, it requires the activation of both the structural state and the cognitive state. In this context, the brain processes external images, activates phonological and structural states, and integrates them within the Language Organ, thereby forming the language state.

4.6 The Significance of the Theory of New Language State in the Study of Chinese Characters

The significance of the Theory of New Language State in the study of Chinese characters lies not only in its profound understanding of the functional roles of Chinese symbols but also in further clarifying the unique position of Chinese characters in cross-linguistic research. By deeply exploring the integrated structure of "form-sound-meaning" in Chinese characters, the theory breaks through traditional research frameworks and offers a novel perspective on understanding and comparing the construction and evolution processes of different written symbol systems.

The Theory of New Language State reveals the multifaceted identity of Chinese characters in the language system through the concept of "Language State". Unlike traditional linguistic theories that emphasize either the independence of character forms or the functionality of words, the Language State Theory highlights the dynamic roles of Chinese characters in language storage, generation, and transmission. By combining structural state (character form) with cognitive state (linguistic meaning), Chinese characters are not merely language symbols but also carriers of linguistic structures. The morphological structure of Chinese characters directly reflects the morphemes or meanings they represent, while simultaneously activating related contexts and concepts at the cognitive level. This unique characteristic allows Chinese characters to efficiently convey information across various contexts, enabling them to play multiple roles within the Chinese language system—not only as building blocks of vocabulary units but also as effective tools for semantic transmission and cultural expression.

Traditional theories, such as "character-based" and "word-based" approaches, have analyzed the linguistic functions of Chinese characters from different angles but have overly focused on single dimensions, making it difficult to explain the comprehensive roles of Chinese characters in the language system. The Theory of New Language State overcomes this limitation by providing a multidimensional analytical framework for the study of Chinese characters. Through the Language

State Mechanism, this theory integrates the generation mechanism (the formation of Chinese characters) and the storage mechanism (the role of Chinese characters in memory), offering explanations for the multifunctionality of Chinese characters from both cognitive and cultural perspectives. This perspective not only explores the functions of Chinese characters at the grammatical and lexical levels but also provides a deeper understanding of their roles in language generation and storage, revealing their adaptability across different linguistic contexts.

Compared with the linear structure of alphabetic writing systems, the distinctive feature of Chinese characters lies in their non-linear characteristics. By constructing relationships between shapes and morphemes, Chinese characters integrate visual symbols with phonetics and semantics to form a unified "form-sound-meaning" structure. The Theory of New Language State further unveils the cognitive and linguistic mechanisms behind this non-linear characteristic. Chinese characters trigger cognitive responses through their visual forms during the transmission of linguistic information, achieving both immediate language expression and deep semantic accumulation and storage. This non-linear structure allows Chinese characters to carry more cultural and historical information and to be flexibly used across various contexts, reflecting the richness and adaptability accumulated through long-term language development.

The Theory of New Language State not only contributes to internal research on Chinese characters but also provides new theoretical support for comparative studies of cross-linguistic symbol systems. Traditional linguistic research has primarily focused on the symbol systems of individual languages. In contrast, the core concept of "Language State" in the Theory of New Language State emphasizes the universal mechanisms of generation and storage behind language symbols. This perspective enables deeper comparisons between Chinese characters and other written symbol systems, such as alphabetic and phonetic scripts. The "form-sound-meaning" integrated structure of Chinese characters contrasts sharply with the separation inherent in phonetic writing systems. This difference not only highlights two distinct writing traditions but also provides new insights into understanding the generative principles of different symbol systems. For example, while phonetic scripts form language symbols through the combination of letters and phonemes, Chinese characters achieve language expression through the direct integration of shapes and morphemes. This structural difference reflects the varying logics of language generation in different symbol systems, revealing the diverse functions of written symbols in linguistic expression. The Theory of New Language State offers a universal analytical framework to help us understand how different language symbol systems achieve the storage and transmission of linguistic information through diverse means.

Beyond linguistics, the Theory of New Language State also offers valuable insights into artificial intelligence (AI), particularly in the field of natural language processing (NLP). In technical areas such as machine translation, speech recognition, and intelligent writing, the conversion mechanisms between Chinese characters and phonetic scripts have long been a research focus. By analyzing the semiotic characteristics of Chinese characters, the theory proposes a language generation mechanism based on the "form-sound-meaning" integration, providing a new theoretical framework for transforming Chinese characters into phonetic scripts. This framework helps improve the accuracy of machine translation systems, especially in mapping relationships between Chinese characters and other languages, better capturing the complexity and diversity of language structures.

The Theory of New Language State also promotes cross-cultural linguistic research. By comparing Chinese characters with other language symbol systems, researchers can gain deeper insights into the cognitive patterns behind language expressions in different cultural contexts. For example, the iconicity and pictographic features of Chinese characters contrast with the abstract nature of alphabetic scripts, revealing different thinking modes in language expression across cultures. This research not only facilitates cross-cultural understanding but also provides theoretical support for cultural exchanges in multilingual environments. Through the lens of the Theory of New Language State, we can better comprehend the cultural logic and cognitive patterns underlying language, offering new solutions for promoting linguistic diversity and cultural integration.

In conclusion, the Theory of New Language State has far-reaching implications for the study of Chinese characters. It not only breaks through traditional theoretical frameworks but also provides a new theoretical foundation and perspective for in-depth research in areas such as linguistics, cross-linguistic studies, and AI applications. The introduction of this theory enhances our understanding of the complexity of Chinese characters and offers valuable insights into exploring the commonalities and differences among various language systems.

5. Conclusion

Professor Lu Jianming once remarked to the author: "Proposing new ideas and seeking new explanations is good, but it is essential to ensure that these ideas and explanations are substantive, rather than merely introducing new terminology." Inspired by this advice from Professor Lu, the author has deeply reflected on the question, "What problems does the New Language State Theory solve?"

The first issue addressed is the distinction between "character" and "word" in Chinese linguistics. Within the framework of the Language State Generation Mechanism, we have clarified why there are different perspectives on "characters" in Zhao Yuanren's sociological and linguistic studies, as well as on "words." The reason is that the discussions often stand on opposite sides of the language state (i.e., focusing on either the structural state or the cognitive state), which has led to confusion between the sociological "character" and the linguistic "character" (i.e., structural state). The author aims to elucidate this issue.

The second problem addressed concerns the existing concept of "character" and the foundational theories of Chinese script. The author agrees with Professor Qiu's analysis that the current definition of "character" conflates two distinct concepts—writing as a physical form and linguistic encoding as storage. This dual nature complicates the explanation of concepts related to decoding and reverse transmission in the Language Storage-Transfer Mechanism. Although scholars in ancient Chinese philology have begun distinguishing between "character (structural state)" and "word (language state)," there remains ambiguity in understanding the essence of Chinese script and Chinese linguistics. The New Language State Theory provides a universal linguistic framework for the study of Chinese characters, as it is applicable to any written symbol system. This theory resolves fundamental issues in the study of scripts and facilitates the advancement of Chinese script research towards a more universal study of written symbols (or universal language state theory).

In the study of Chinese linguistic history and philology, the author has also applied the Language State approach, achieving what Professor Lu recommended—offering new ideas and explanations that are substantive. The longstanding debate between "character-centric" and "word-centric" approaches has inspired the author to contribute a new perspective for scholarly discussion. The Language State approach represents an exploratory step towards integrating Chinese philology with the broader study of written symbols.

As Shi Feng(2023)advocated, we must "reread Zhao Yuanren," because "re-examining Zhao Yuanren is, in essence, re-examining ourselves, allowing us to see the distance between our understanding and his." Shi Feng cites Wu Zongji’s comment: "Many of his arguments were groundbreaking for their time, making them difficult to understand. His insights into linguistics were almost prophetic..." Through an in-depth analysis of Zhao Yuanren’s views on characters and words, this paper reveals the unique role and essential features of Chinese characters within the Chinese language. Zhao’s perspectives not only offer new insights for the study of Chinese characters but also lay a solid foundation for modern Chinese linguistic research.

The primary contribution of this study lies in systematically exploring the unique functions of Chinese characters through the New Language State Theory, thereby enriching the theoretical framework for Chinese character research.

Future research can delve deeper into the following areas: First, incorporating more empirical language data to test the applicability and limitations of Zhao Yuanren's views on characters and words; second, investigating the application of Zhao's perspectives in modern Chinese language education, with the aim of improving the scientific rigor and effectiveness of Chinese character instruction; and third, conducting cross-linguistic comparative studies to explore the uniqueness and universality of Chinese characters across different language systems.

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